Subashanan Nair

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 7+ years of experience across machine learning, data engineering and technical education. Builds production data pipelines, dashboards and LLM powered tools for Penang Institute and Forward College. Recent published work covers Malaysian public statistics: a road distance study of access to services in Penang, leakage safe traffic forecasting, and evaluations of LLMs on out of date open data and on governed text to SQL. Hands on with Python, scikit-learn, TensorFlow, Streamlit, Docker and GitHub Actions. DataCamp certified AI Engineer for Data Scientists and Kaggle Top 100 competitor.

SKILLS

Languages	Python, SQL, R, JavaScript, TypeScript
ML & AI	scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, CatBoost, Deep Learning, Reinforcement Learning, LLM applications (Ollama, prompt design, RAG, text to SQL)
Data & MLOps	Pandas, NumPy, SciPy, Streamlit, Plotly, Matplotlib, Seaborn, MLflow, Docker, GitHub Actions, Tableau, Google Drive API
Web & Databases	Next.js, FastAPI, Node.js, REST API design, PostgreSQL, Supabase, Quarto
Tools	Git, GitHub, Jupyter Notebook, Streamlit Cloud

EXPERIENCE

Forward College | *Instructor & Data Scientist* Feb 2019 to Present

- Instruct 3 cohorts per year in Python, machine learning and data analytics, averaging a 4.7/5 learner satisfaction score.
- Built internal programme tools: an LLM assisted project review dashboard, a GitHub based submission and assessment platform, and an exam monitoring app.
- Engineered and deployed ML pipelines and predictive models with Python, scikit-learn and MLflow for client organisations, reducing manual analysis time by 60%.
- Delivered data solutions and dashboards for 3+ organisations, including Penang Institute, enabling data driven policy decisions.

Penang Institute | *Data Scientist (Consultant)* Jun 2024 to Present

- Diagnosed a stale public statistics site and replaced a 34 script R workflow with an automated Python pipeline (GitHub Actions, Docker) that publishes OpenDOSM and data.gov.my data to Google Drive for Tableau.
- Built a weekly ingestion pipeline for key Penang indicators (population, GDP, employment, CPI, household income) backed by a 31 test validation suite.
- Created a pull request gated Streamlit uploader so staff can add new publications without touching code, and an interactive key statistics dashboard (Streamlit, Plotly) deployed on Streamlit Cloud.
- Maintained the Quarto statistics website and delivered a client requested website revamp.

Kuen Lai International Secondary School | *STEAM Consultant* May 2013 to Jan 2019

- Taught Python programming, Arduino and Raspberry Pi electronics, and 3D printing through hands-on, project based STEAM lessons.
- Designed and delivered STEAM programmes aligned with national and international curriculum frameworks.
- Ran workshops for educators on STEAM pedagogy and digital literacy.

PROJECTS

Penang Pulse | *Spatial Analysis*

Python, NetworkX, rasterio, Shapely, OpenStreetMap, WorldPop, DOSM data, Next.js

- Reproducible study of how far Penang residents live by road from public clinics, schools and Rapid Penang stops. 20% of Seberang Perai Selatan residents live more than 3 km from a public clinic (21% with Meta's population map) and 74% of Penang residents live within 1 km of a Rapid Penang stop, across 2,437 facilities and stops mapped. Limitations are published with every result and artefacts carry SHA-256 checksums.

Forecasting Lab | *Predictive Modelling*

Python, scikit-learn, SciPy, PyTorch, Chronos, UCI dataset 608

- Leakage safe traffic forecasting benchmark across a 36 sensor network, comparing five models from 15 to 120 minutes ahead with block bootstrap intervals. Gradient boosting cut MAE by 23% versus persistence at 15 minutes (95% interval 16 to 30%); zero shot Chronos beat it by 26% at 2 hours (17 to 34%) and had the lowest error at every horizon.

LLM Eval Lab | *LLM Evaluation*

Python, Ollama Cloud, tool calling, PyMuPDF, bootstrap intervals, data.gov.my, DOSM, BNM, MYNIC

- Tested whether cloud LLMs find Malaysia's latest official figure when open data lags the release: 30 DOSM, BNM and MYNIC statistics, 120 questions verified against official releases. On out of date series with tools, DeepSeek V4.1 Flash gave the latest figure for 91% (95% interval 78 to 100%) and gpt-oss 120B for 88% (77 to 97%), against 0% for the open data alone and 0% from memory.

Penang Economic Data Assistant | *Prototype, Evaluated*

FastAPI, PostgreSQL, Next.js, Docker, Ollama Cloud (gpt-oss:120b)

- Natural language assistant over OpenDOSM CPI and unemployment data. Questions are translated to governed SQL with an audit trail, alongside a Next.js data explorer. Passed all 6 release gates over 45 cases run 3 times each: 81 of 81 supported answers correct, 45 of 45 critical scope cases handled, median latency 1.6 s.

Forward College Project AI Review Tool | *Internal Tool*

Python, Streamlit, GitHub API, Ollama Cloud (gpt-oss:120b), GitHub Actions

- Monitors 9 student capstone repositories and scores each against a 10 section rubric. The LLM reviewer must cite exact source quotes as evidence; fabricated quotes are rejected and the final percentage is computed locally, not taken from the model. Runs hourly on GitHub Actions and never executes student code.

FC Submission Platform | *Internal Tool*

Next.js, TypeScript, Supabase, GitHub OAuth, Playwright

- Students register a public GitHub repo; the platform snapshots the commit at the deadline and runs an automated deliverable checklist. Instructors can override results and grant exceptions with a full audit trail. LLM rubric grading is specified for the next phase.

Exam Guard, Exam Monitoring App | *Internal Tool*

Rust, Tauri, TCP/UDP networking, GitHub Actions

- LAN proctoring tool needing no accounts or internet: a live grid of student screens with problem tiles surfaced first, UDP discovery and class codes. Ships cross platform releases for Windows, macOS and Linux.

Documentation MCP Server

JavaScript, Node.js, Docker, REST API

- Aggregates live documentation from open source libraries (React, Vue, Angular and more) into a searchable REST API and web UI, with daily cron updates and Docker Compose deployment.

EDUCATION

Asia e-University | *Master of Education in Science* — May 2014 to Oct 2016

Universiti Tun Abdul Razak | *Bachelor of Education (Hons)* — May 2009 to Sep 2012

CERTIFICATIONS

DataCamp | *AI Engineer for Data Scientists Associate* — Apr 2026

Certificate ID: AEDS0019212905886

Coursera (DeepLearning.AI & Stanford) | *Supervised ML: Regression and Classification* — 2024

Google for Education | *Google Certified Educator Level 1* — Dec 2021

Kaggle Competition | *Top 100 placement in a global machine learning competition*